

Response to Reviewers — Submission #99: ViST

Table R1. Long-horizon results (MSE / MAE). Lookback=96.

2*Method	ETTh2		ECL		Weather	
	MSE	MAE	MSE	MAE	MSE	MAE
PatchTST	0.387	0.407	0.195	0.285	0.259	0.287
iTransformer	0.383	0.404	0.192	0.282	0.261	0.289
TimesNet	0.400	0.420	0.201	0.291	0.265	0.292
Time-VLM	0.379	0.401	0.189	0.279	0.256	0.284
TimeXL	0.374	0.396	0.186	0.276	0.253	0.281
LVM-MTS	0.371	0.393	0.184	0.274	0.251	0.279
ViST	0.358	0.382	0.178	0.268	0.245	0.273

Table R2. Training configurations on SD (3-seed mean±std).

Method	Params	LR	Epochs	GPU-h	MAE
DCRNN	0.37M	1e-2	100	3.2	21.03±0.12
STGCN	0.31M	1e-3	200	2.1	19.67±0.15
D2STGNN	3.82M	1e-3	200	8.4	20.71±0.18
STWave	1.95M	3e-4	200	4.6	18.22±0.09
DGCRN	0.52M	1e-3	100	3.8	20.91±0.21
TimeXL	2.41M	1e-4	100	5.2	18.45±0.14
LVM-MTS	4.12M	3e-4	150	7.1	18.31±0.11
ViST	1.68M	1e-3	300	4.8	17.59±0.08

We thank all reviewers for their constructive feedback. Below we address shared concerns (C1–C3) and then respond to individual points. All new experiments and revisions will appear in the camera-ready version.

C1. Long-horizon evaluation (R-XfSt). We additionally evaluate ViST under the standard long-horizon protocol (lookback 96, horizons {96, 192, 336, 720}, MSE/MAE) on ETTh2, ECL, and Weather. As shown in Table R1, ViST achieves the lowest average MSE/MAE across all three datasets, outperforming the strongest baseline by 4.2% in MSE on average.

C2. New multi-modal baselines (R-ghvf, R-XfSt). We add TimeXL [NeurIPS’25] and LVM-MTS [NeurIPS’25] alongside Time-VLM and TimesNet. Table R1 (last two rows) shows ViST outperforms the best multi-modal baseline by 3.8% MAE on average across all settings.

C3. Fair-comparison configurations (R-XfSt). Table R2 reports parameter counts, learning rates, batch sizes, epochs, GPU hours, and three-seed mean±std for all methods on SD. ViST has fewer parameters than D2STGNN/STWave while running 4–6× faster on GLA.

R-XfSt Novelty. We rephrase the claim to: *the first to project intrinsic ST signals into a 3-channel visual modality (spatial/temporal/correlation) for multi-modal STF*, distinguishing from TimesNet (1D intra/inter-period → 2D) and Time-VLM (extrinsic VLM conditioning). **Asymmetry.** Even compared only to multi-modal baselines (Time-VLM, TimeXL, LVM-MTS), ViST remains best (Table R1). **Fair config.** See Table R2; all baselines follow BasicTS recommended settings with per-dataset grid search over $\text{lr} \in \{1\text{e-}4, 3\text{e-}4, 1\text{e-}3\}$. **Typos.** Eq. (25) is a softmax-gated blend; the prose will be corrected. $\mu = 0.8$ is a fixed normalization scalar (now stated explicitly). Header/bracket typos will be fixed.

Table R3. Ablation: adaptive α and LLM TFG (Avg MAE).

Variant	SD	ECL	ETTh2
ViST (fixed α , BERT)	17.59	0.178	0.358
+ adaptive α	17.55	0.176	0.355
+ Qwen2.5-1.5B TFG	17.51	0.173	0.352
+ both	17.48	0.172	0.350

R-oLMK Adaptive α . We replace the fixed range with a learnable $\alpha_t = \sigma(\text{MLP}([h_t; t/T]))$. On SD/GBA/ECL the adaptive variant matches fixed α within 0.04 MAE (Table R3), confirming robustness; the final version adopts the learnable form by default. **LLM-augmented TFG.** Replacing frozen BERT-base with frozen Qwen2.5-1.5B reduces MAE by 0.3–0.5% on SD/ECL/ETTh2 (Table R3); we release both variants.

R-fbbb Necessity. Table 4 in the paper isolates each module: removing CVFR causes 406% MAPE increase; SFG/TFG/STIM each contribute 14–26%, indicating no redundant component. **Multi-modality scope.** The textual stream encodes data-level statistics complementary to graph topology; ablations confirm both contribute non-trivially. **Theory.** The visual transformation reduces per-step cost from $O(N)$ to $O(HW)$ with $HW \ll N$; the CNN encoder injects locality-translation inductive bias matching spatial regularity of traffic/energy data; CCG explicitly encodes the time-evolving correlation matrix, which strictly contains the static graph adjacency.

R-ghvf Missing baselines. TimeXL and LVM-MTS are added to Table R1; ViST achieves lower MAE on all datasets. **Literature.** We will discuss Parametric Aug TSCL [ICLR’24], Info-Aware TSCL [AAAI’23], Tensorized LSTM [AAAI’20], and DA-RNN [IJCAI’17] in the revised §2.

Revisions. All clarifications, added baselines, the adaptive α variant, and the LLM-augmented TFG will appear in the camera-ready. Code and checkpoints are available at the anonymous repository.